

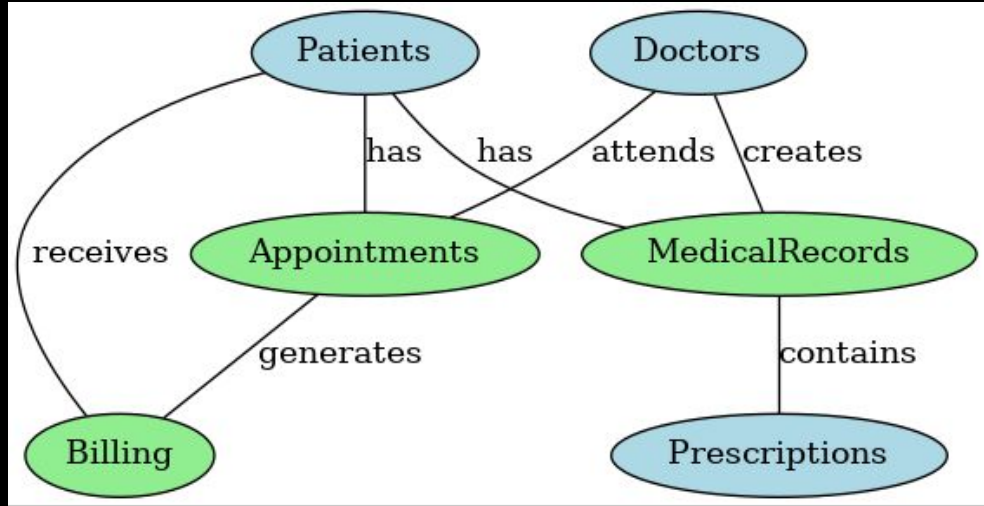
Diversity Healthcare



Fall 2025

The problem

Functionality and Information is Disconnected



Digital health data is scattered across disconnected systems, causing delays, errors, and unequal access to care. Clinicians must navigate multiple platforms for records and imaging, while patients lack a simple way to view and manage their own information.



Our approach



Integrating scheduling medical records and prescriptions into one seamless mobile application





Why It Matters

- Fragmentation reduces efficiency, increases errors, and widens health inequities.
- A single, usable system could improve continuity of care.



Literature Review: Fragmentation

- Studies show widespread usability and accessibility failures in health systems.
- Disjointed EHRs burden users and disrupt workflows.



Project Goal

- Design a patient-centered, accessible, unified health application.
- Supports appointments, records, and prescriptions in one place.



Literature Review: Patient Outcomes

- Unified access improves engagement, adherence, and confidence.
- But poor design excludes disabled users and low-literacy groups.



Opportunity Areas

Three core gaps:

1. Integration
2. Accessibility
3. Role-Differentiation

Addressing these can reduce fragmentation and enhance usability.

Research Question

How can we design a unified, accessible digital health platform?

And how can it support both clinicians and patients effectively?

Gathering User Requirements



Approach: User-Centred Design

Techniques Used

- Surveys
 - 29 Patient Responses
 - Easy Distribution
 - Quick Responses
 - Simpler Data Analysis
- Semi-structured Interviews
 - Doctors, Medical & Clinical Staff
 - 11 Interviews
 - Accurate & High Quality responses
 - Guided by Script with Flexibility

Preferences & Accessibility

What is your preferred sign-in method on secured mobile apps? *

☐ FaceID/TouchID to unlock; password only on first login

☐ 6-digit app PIN to unlock; password only on first login

☐ Password everytime

What accessibility options should a healthcare app have? *

☐ Adjustable text size (Larger texts)

☐ High contrast/dark mode

☐ Screen reader support (auditory support)

☐ Adjustable button size (Larger buttons)

☐ None

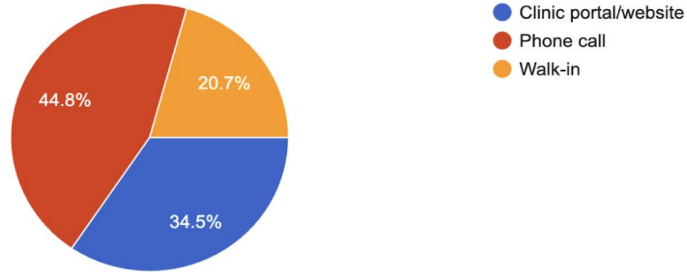
☐ Other: _____

Back Submit Clear form

User Requirement Results (Surveys)

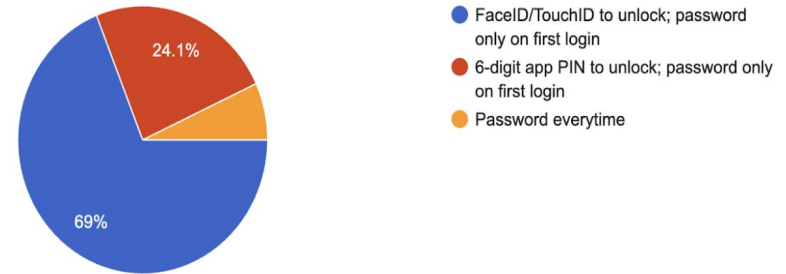
How do you book appointments now?

29 responses



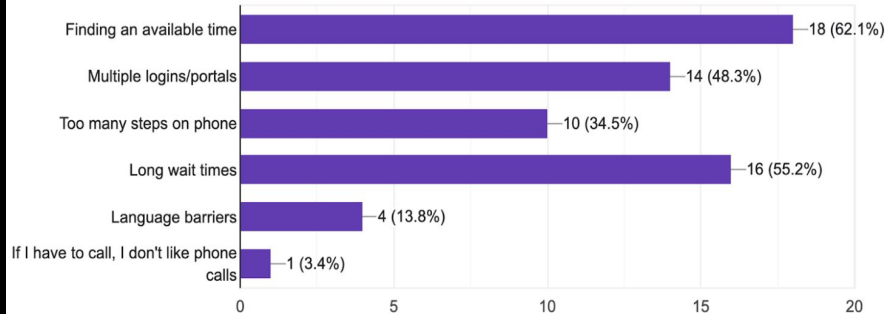
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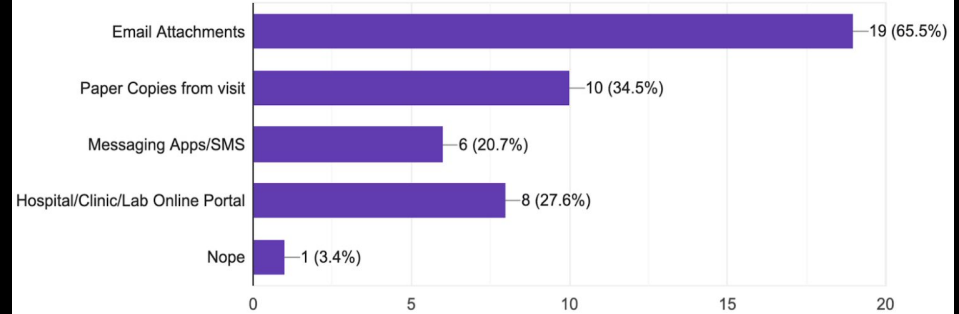
What are the biggest struggles in your current booking process?

29 responses



How do you currently receive health records and new results?

29 responses



User Requirement Results (Interviews)

From the discussions, the following **key challenges** emerged:

Challenge	% of students mentioning it	Example insight
Fragmented patient information (multiple systems, poor data flow)	10/11	“We can’t see a patient’s full history easily.”
Time wasted on paperwork or retrieving data	9/11	“Documentation takes so long; we just skip it sometimes.”
Poor accessibility or usability of existing tools	8/11	“Apps are too cluttered or slow on mobile.”
Lack of secure and easy data sharing among peers/doctors	7/11	“We use screenshots to share lab reports — that’s not safe.”
Limited offline access	5/11	“Wi-Fi is patchy in wards. An offline mode would be great.”

Interactive Prototype

- Figma Prototype
- 3 Main Tasks:
 - a. View Clinics Around the Area
 - b. View/Book/Cancel Appointments
 - c. View and Create Patient Records and Prescription Records



Interactive Prototype

Task 1: View Clinics Around the Area

Patients:

- Scan a distance-sorted list of nearby clinics
- Filter or search to narrow options fast (e.g. Open Now, Favourites, Languages)
- View a clinic's details (e.g. hours, services, languages, map location, etc)

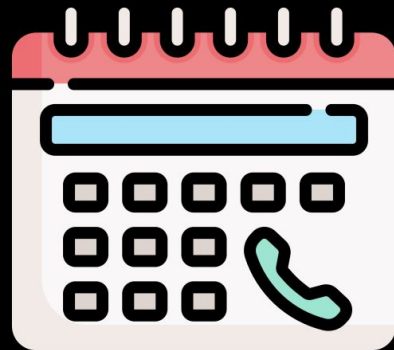


Interactive Prototype

Task 2: View/Book/Cancel Appointments

Patients:

- Browse nearby clinics and available doctors/time slots
- Book an appointment with a chosen clinic, provider, date, and time
- View upcoming appointments in one place
- Edit time/date or cancel and rebook when needed



Interactive Prototype

Task 3: View and Create Patient Records and Prescription Records

Patients:

- Review a list of health records with filters and search function available
- Open a record to see a clear summary and detailed result view
- View their prescription orders and nearby pharmacies for pickup

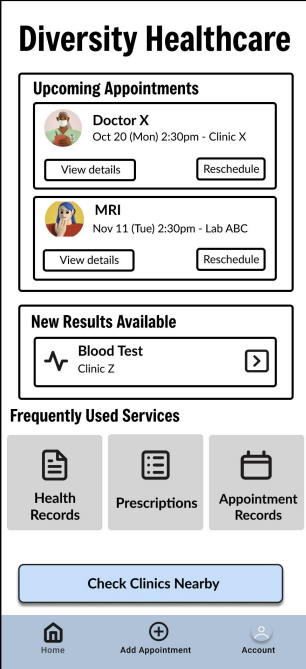
Doctors:

- Add new patient health records (text details + file uploads)
- View and update existing patient records and prescriptions

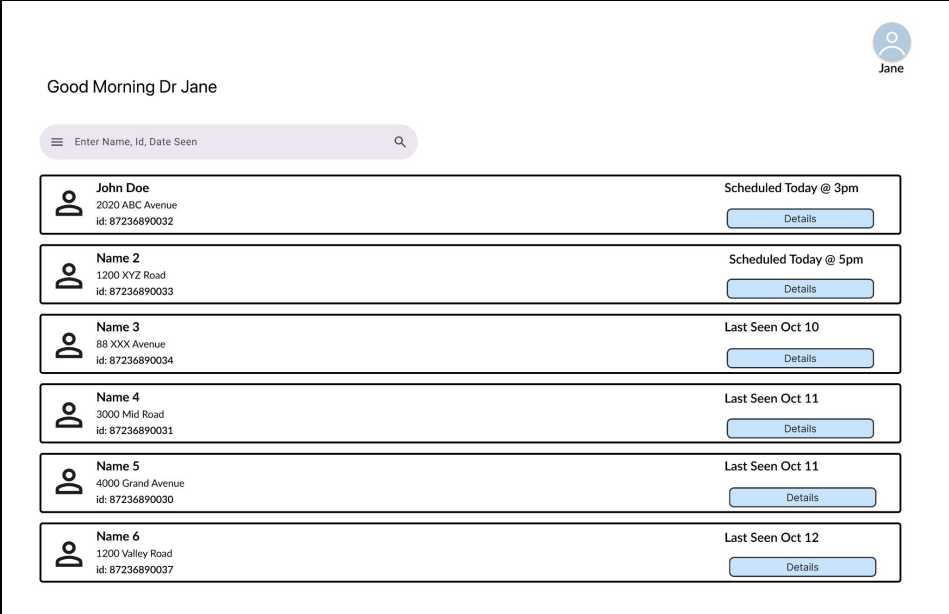


Figma Prototype Demo

Patient Flow: Tasks 1+2+3



Doctor Flow: Task 3



Usability Testing

Questionnaires

- Primary data-collection method for effectiveness
- The survey is divided into four categories
- Remote Environment
- Required consent form



Google Forms

1. Personal Background

- Age
- Relationship with healthcare service(dr/nurse/clinical staff/patient)
- If experienced digital healthcare system before
- Frequency of healthcare visits

What is your relationship to healthcare services in real life? *

- ☐ I am a patient
- ☐ I am a doctor / physician.
- ☐ I am a nurse or other clinical staff.
- ☐ I work in healthcare administration (reception, records, etc.).

2. Patient Flow Feedback

- Booking an appointment was easy
- Editing/Canceling an appointment was easy
- It was easy to find my health records
- Test results were understandable
- The clinic list was easy to browse and provided clear detailed

Each question in this section
was delivered in a scale-based response format

Test results were understandable

1	2	3	4	5
☆	☆	☆	☆	☆

3. HealthCare Worker Flow Feedback

- Adding a new patient record was intuitive
- Adding or updating a prescription order was straightforward
- Updating an existing record was easy

Each question in this section was delivered in a scale-based response format

Adding or updating a prescription order was straightforward



4.Satisfaction

- What features do u like the most about our application
- Is there anything you were dissatisfied with while using this app? If yes, please describe

what feature do u like the most about our application *

☐ minimalistic design

☐ overall usability

☐ record accessibility

☐ search features(clinic)

☐ book/edit/cancel appointments

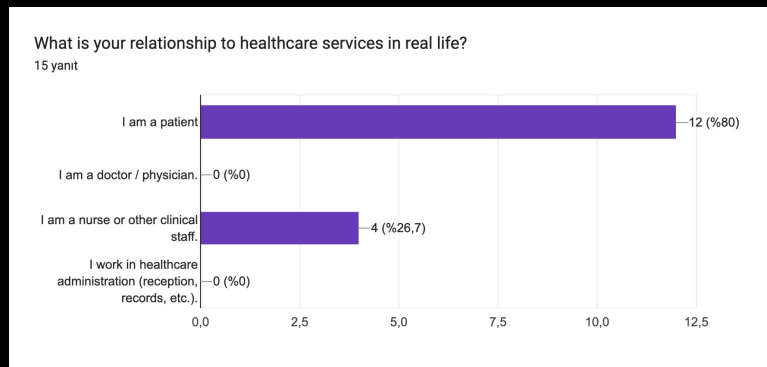
☐ readability

☐ Other: _____

Questionnaire Results

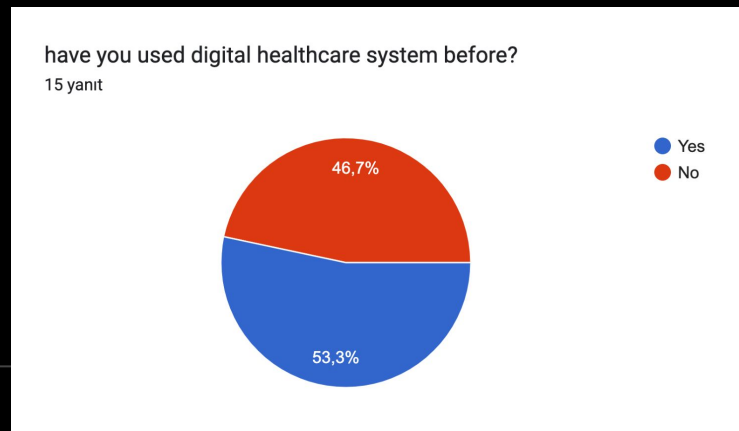
Participants

- 15 total
- Ages 20–50
- 80% patients, 27% nurses
- 53% used digital healthcare apps before



Goals

- Evaluate booking, records, prescriptions, and clinic browsing
- Identify usability pain points
- Measure satisfaction



Appointment Booking:

Ratings:

- Booking an appointment: **3.93 / 5 -> 78.6%**
- Editing/canceling: **4.00 / 5 -> 80%**
- Clinic list clarity: **3.93 / 5 -> 78.6%**

Health Records:

Ratings:

- Finding records: **4.08 / 5 -> 81.6%**
- Understanding test results: **3.86 / 5 -> 77.2%**

Strengths

- Clear booking flow
- Easy to modify appointments
- Clinic details useful
- Clean Layout, scrollable and familiar format

Issues

- Some buttons misfired
- Same labels were unclear
- Filters occasionally behaved incorrectly

Doctor/Nurse Admin Tools

Ratings:

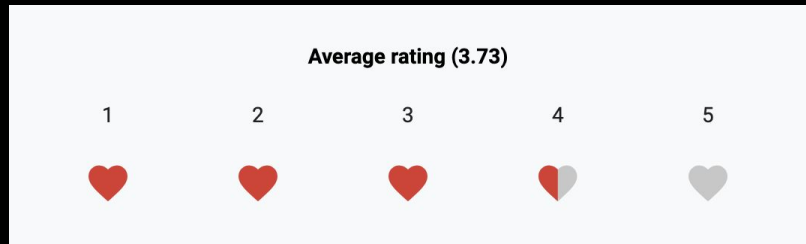
- Adding patient record: **3.64 / 5 -> 72.8%**
- Updating prescription: **3.73 / 5 -> 74.6%**
- Editing existing records: **3.82 / 5 -> 76.4%**

Strengths

- Logical Layout
- Clear separation between prescriptions and records

Issues

- Some buttons misfired
- No reassurance of the updates



Overall Results

Strengths

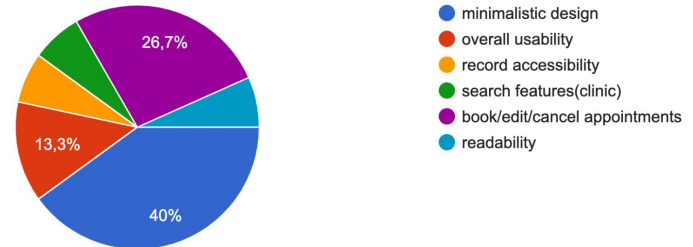
- Good overall usability (scores ~73%–82%)
- Strong core workflows
- Clean, minimalistic visual design
- Users easily found main features

Weaknesses

- Interaction reliability issues
- Button inconsistencies
- Inadequate feedback when saving/editing

what feature do u like the most about our application

15 yanıt

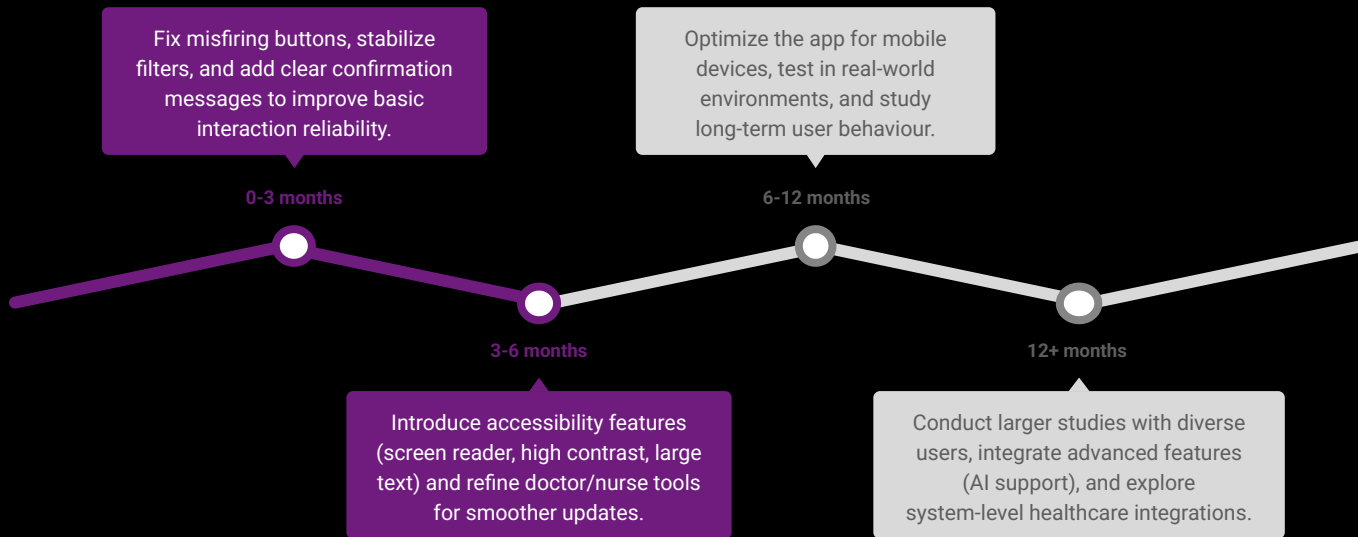


Limitations

- Limited participant diversity: Few older adults, low-literacy users, or people with disabilities resulting in non generalised results
- Prototype limitations: Mid-fidelity prototype caused issues like misfiring buttons and inconsistent filters.
- Short task-based study: Only tested one-time tasks; long-term usage patterns not captured.
- No accessibility testing: Screen readers, large text, and high-contrast modes were not evaluated
- Limited device testing: Focused on desktop/mobile; not tested on medium/large screen tablets or real-world environments.
- Small clinician sample: Few nurses/doctors participated, limiting insights for professional workflow.



Future Work & Improvements



Thank You!



Fall 2025



References

- Phase 1 Documentation
- Phase 2 Documentation
- Phase 3 Documentation
- Phase 3 Prototype, Prototype Patient Flow, Prototype Doctor Flow
- google form